

Arush Jain

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Summary

B.Tech student specializing in Smart Manufacturing with a strong foundation in Machine Learning, LLM Systems, and Data Structures & Algorithms. Experienced in designing, training, and deploying end-to-end AI systems – including multi-agent LLM pipelines with tool-calling and reflection loops, hybrid retrieval-augmented generation (RAG) systems with cross-encoder reranking and statutory-grounded citations, and deep learning recommender systems with FAISS-based vector retrieval – deployed on AWS and GCP and complemented by full-stack engineering skills (React.js, Next.js, Node.js) for shipping production-grade APIs. Proven ability to independently architect, benchmark, and ship AI-driven products, backed by rigorous quantitative evaluation against baselines and controlled ablation studies. Seeking to apply strong LLM engineering and ML fundamentals to build reliable, production-grade AI systems.

Education

IIITDM Jabalpur
B.Tech in Smart Manufacturing

2023 – Present

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, C++

Generative AI & Agentic Systems: LLM APIs (Google Gemini, Cohere), Retrieval-Augmented Generation (RAG), Hybrid Search (FAISS + BM25), Cross-Encoder Re-Ranking, Multi-Agent Orchestration & Reflection Loops, Function/Tool Calling, Prompt Engineering, AWS Textract (Document AI/OCR)

Machine Learning: PyTorch, TensorFlow/Keras, LightGBM, scikit-learn, Pandas, NumPy, Conformal Prediction, Quantile Regression, Contrastive Learning (InfoNCE), Two-Tower / Dual-Encoder Retrieval, Recommender Systems, Time-Series Forecasting

MLOps & Cloud Deployment: FastAPI, Docker, Kubernetes, MLflow, Airflow, CI/CD (GitHub Actions), AWS (S3, Textract), Google Cloud Platform (Compute Engine, Cloud Run), Render, Streamlit Cloud

Web Development: React.js, Next.js, Node.js, Express.js, REST APIs, Streamlit, Tailwind CSS

Databases: MongoDB, PostgreSQL, Supabase, FAISS (Vector Search)

Tools: Git, GitHub, Postman, Figma

Core CS: Data Structures & Algorithms, Probability & Statistics, Object-Oriented Programming, System Design

Projects

RailRouteAgent

Python, Google Gemini, Streamlit

Multi-Agent Reflection System for Split-Journey Route Planning

- Architected a 3-agent (Planner/Verifier/Ranker) system using Gemini function calling to discover operationally-safe split-journey train routes, with a critic-refinement loop that rejects infeasible transfers and re-plans.
- Benchmarked against a zero-shot LLM baseline across 12 test scenarios, raising the operational feasibility pass rate from 0% to 100% and cutting end-to-end latency 2.2x (3.2s → 1.45s) at ~\$0.0006 per search.
- Shipped a live Streamlit application with an execution-trajectory viewer exposing agent prompts, tool calls, and reflection critiques for full pipeline observability.

Chambers & Infrastructure

FastAPI, AWS, Cohere, Gemini, Docker

Hybrid RAG Legal Copilot & Automated Invoice Tax Auditor

- Built a dual-stream hybrid retrieval pipeline (FAISS dense + BM25 sparse, distilled via Cohere Rerank v3.5) over the Indian GST Act to ground legal Q&A with verified statutory section-level citations.
- Automated invoice tax-rate auditing using AWS Textract OCR to extract line items and cross-validate GST rates, deployed as a containerized, API-key-gated FastAPI service on AWS S3 + Render.
- Ran a 20-question statutory benchmark showing hybrid retrieval cut the hallucination rate from 36.8% to 2.1% and raised retrieval Precision@4 from 48.3% to 94.2% over naive vector search.

Multimodal Two-Tower Product Recommender

PyTorch, FAISS, scikit-learn

Deep Learning Candidate Retrieval System with Hard-Negative Mining

- Designed and trained a two-tower dual-encoder retrieval model (user/item MLP towers with FAISS similarity search), benchmarked on a MovieLens warm-start split spanning 55K+ users and 20K+ candidate items.
- Implemented a hard-negative InfoNCE contrastive training loop with a custom FAISS-based hard-negative sampler, improving Recall@10 by 4.2% and Recall@50 by 3.6% over an in-batch-negative baseline.
- Ran a controlled 6-model ablation study (text features, gated fusion, Stage-2 reranking) to isolate architectural impact, backed by a 53-test pytest suite guaranteeing zero train/test leakage.

Patent

Automatic Footrest Assembly for Two Wheeler

2025

- Designed a sensor-based automation system using embedded controllers and pressure sensors.
- Implemented hardware-software integration workflows with fail-safe operational logic.

Achievements

- Google Cloud Arcade Facilitator Program** - deployed cloud-native applications on GCP using Compute Engine and Cloud Run.
- Selected for **Amazon ML Summer School 2026 (AMLSS)**, a competitive program focused on Machine Learning and Applied Data Science.

Co-Curricular Activities

- Organized and managed College Literature Club events, coordinating execution, logistics, and team workflows.
- Active member of a community Run Club, co-organizing running events and group games.